API Documentation

HTTP API reference: search, recommendations, itinerary feasibility, and feedback

Version	Not specified
Date	Not specified
Base URL	`http://127.0.0.1:8000`
API prefix	`/api/v1`
Content type	`application/json`
OpenAPI	`/docs` and `/openapi.json`

Compiled only from the uploaded API_Documentation.md. Sample values are illustrative where noted.

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1. Overview

1.1 Connection details

Item	Value
Base URL	`http://127.0.0.1:8000`
API prefix	`/api/v1` (used by search, recommendations, itinerary feasibility, and feedback; `/health` is documented without it)
Request content type	`application/json` for JSON requests
Responses	JSON for all successful endpoint responses
Interactive OpenAPI documentation	`/docs` (includes an interactive request runner)
OpenAPI schema	`/openapi.json`
Authentication	Not specified

1.2 Documented contract versus illustrative values

Field names, validation constraints, status codes, and the behaviors stated for each endpoint form the documented contract. Sample values are examples only.

Documented contract	Illustrative or variable
Field names and validation limits in each request table.	`score`, `keyword_rank`, and `semantic_rank` in samples. They depend on the query and embedding backend.
HTTP status codes shown per endpoint and in Section 4.	`confidence` and the ratings and ranks in sample recommendations. They may vary with the configured dataset and embedding backend.
A valid search with no matches returns HTTP 200 with `count: 0` and an empty `results` array.	`dataset_records` in the health sample. It reflects the configured CSV and can change if another dataset is used.
`status` on feasibility is `insufficient_data` when a budget or duration constraint is provided, otherwise `historical_evidence_only`.	Record field values in samples. They reflect the provided sample row.
Feedback is captured but does not currently change retrieval ranking.	`embedding_backend` is normally `sentence-transformers` or `tfidf`.

1.3 How a request flows

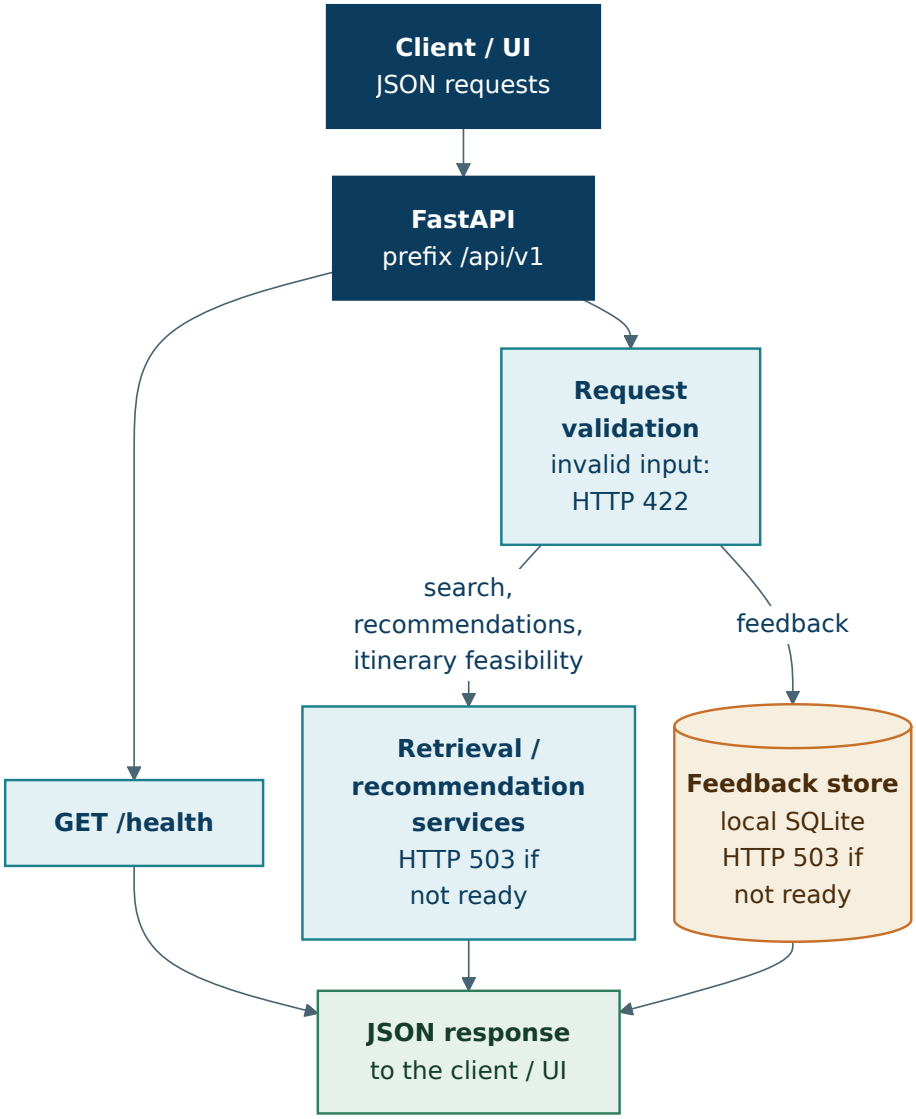


Figure 1. API overview, limited to components and flows named in the API documentation.

Each element in Figure 1 comes from the API documentation:

- **Client / UI** sends JSON requests (`application/json`) to FastAPI at the base URL, using the `/api/v1` prefix for search, recommendations, itinerary feasibility, and feedback.
- `GET /health` checks service readiness, loaded record count, and the active embedding backend.
- **Request validation** rejects invalid input with HTTP 422.
- **Retrieval / recommendation services** perform hybrid keyword/vector search with metadata filters, preference classification, ranking, and the data-coverage assessment for itinerary feasibility. HTTP 503 is returned when the service is not ready.
- **Feedback store** is a local SQLite store. Feedback is captured but does not currently change retrieval ranking. HTTP 503 is returned when the store is not ready.
- **JSON response** is returned for every successful endpoint.

2. Endpoint Map

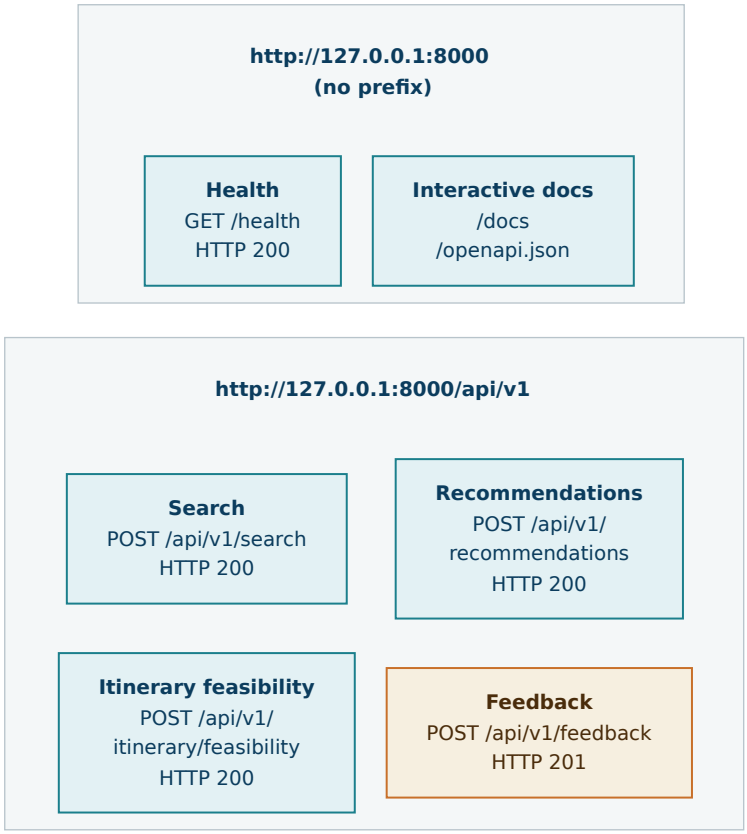


Figure 2. Endpoint map grouped by function, with the documented success status of each endpoint.

Group	Method	Path	Success status	Purpose
Health	GET	`/health`	200	Check service readiness, loaded record count, and active embedding backend.
Search	POST	`/api/v1/search`	200	Retrieve matching experience records with hybrid keyword/vector search and optional metadata filters.
Recommendations	POST	`/api/v1/recommendations`	200	Classify query preferences, retrieve matching evidence, and rank destination and activity summaries.
Itinerary feasibility	POST	`/api/v1/itinerary/feasibility`	200	Return recommendations with a data-coverage assessment for requested itinerary constraints.
Feedback	POST	`/api/v1/feedback`	201	Save feedback in the local SQLite store.

3. Endpoint Reference

3.1 GET /health

Method	Path	Success status
GET	`/health`	200

PURPOSE

Check service readiness, loaded record count, and the active embedding backend.

REQUEST

No request body or query parameters.

RESPONSE FIELDS

JSON object.

Field	Type	Description
`status`	string	Service status (sample: `"ok"`).
`dataset_records`	integer	Number of loaded records.
`embedding_backend`	string	Active embedding backend. Normally `sentence-transformers` or `tfidf`.

> **Illustrative:** `dataset_records` reflects the configured CSV and can change if another dataset is used.

SAMPLE REQUEST

```
GET /health HTTP/1.1
Host: 127.0.0.1:8000
```

SAMPLE RESPONSE (HTTP 200)

```
{
  "status": "ok",
  "dataset_records": 5456,
  "embedding_backend": "tfidf"
}
```

3.2 POST /api/v1/search

Method	Path	Success status
POST	`/api/v1/search`	200

PURPOSE

Retrieve matching experience records using hybrid keyword/vector search and optional metadata filters.

REQUEST

JSON object. `query` is required.

Field	Type	Required	Constraints
`query`	string	Yes	1 to 1,000 characters.
`limit`	integer	No	1 to 50. Default: 5.
`filters`	object	No	Optional metadata filters (below).

`filters` **object.** The source lists these fields without stating which are individually required.

Filter field	Type	Constraints
`destination`	string	None stated beyond type.
`travel_type`	string	None stated beyond type.
`min_rating`	number	From 0 to 5. Cannot exceed `max_rating`.
`max_rating`	number	From 0 to 5. Cannot be less than `min_rating`.

RESPONSE FIELDS

JSON object with `query`, `count`, and `results`.

Field	Type	Description
`query`	string	The submitted query.
`count`	integer	Number of results returned.
`results`	array of objects	Matching experience records (fields below).

Result object (`results[]`).

Field	Type (from sample)	Description
`experience_id`	string	Identifier of the experience record (sample: `"User 24"`).
`destination`	string	Destination of the record.
`categories_visited`	array of strings	Activity categories visited.
`overall_rating`	number	Overall rating of the record.
`review_text`	string	Review text of the record.
`travel_type`	string	Travel type of the record.
`visit_date`	string	Visit date (sample: `"2026-02-24"`).
`score`	number	Retrieval score. **Illustrative** in the sample.
`keyword_rank`	integer	Keyword component rank. **Illustrative** in the sample.
`semantic_rank`	integer	Semantic component rank. **Illustrative** in the sample.
`synthetic_enrichment`	boolean	Flag present on every result; `true` in the sample.

A valid search with no matches returns HTTP 200 with `count: 0` and an empty `results` array.

> **Illustrative:** The record fields in the sample reflect the provided sample row. The score and component ranks are illustrative; they depend on the query and embedding backend.

SAMPLE REQUEST

```
{
  "query": "beaches and restaurants in Lisbon, Portugal for a couple",
  "limit": 1,
  "filters": {
    "destination": "Lisbon, Portugal",
    "travel_type": "Couple",
    "min_rating": 2.24,
    "max_rating": 2.24
  }
}
```

SAMPLE RESPONSE (HTTP 200)


```
{
  "query": "beaches and restaurants in Lisbon, Portugal for a couple",
  "count": 1,
  "results": [
    {
      "experience_id": "User 24",
      "destination": "Lisbon, Portugal",
      "categories_visited": [
        "resorts", "beaches", "parks", "theatres", "museums", "malls",
        "zoos", "restaurants", "pubs/bars", "local services",
        "burger/pizza shops", "hotels/other lodgings", "juice bars",
        "art galleries", "dance clubs"
      ],
      "overall_rating": 2.24,
      "review_text": "Had a so-so time in Lisbon. Visited as a couple. I was let down by the resorts.
The beaches were the highlight of the trip. Only recommended with low expectations.",
      "travel_type": "Couple",
      "visit_date": "2026-02-24",
      "score": 0.0328,
      "keyword_rank": 1,
      "semantic_rank": 1,
      "synthetic_enrichment": true
    }
  ]
}
```

3.3 POST /api/v1/recommendations

Method	Path	Success status
POST	`/api/v1/recommendations`	200

PURPOSE

Classify query preferences, retrieve matching evidence, and rank destination and activity summaries.

REQUEST

Same JSON shape and validation rules as `/api/v1/search`.

Field	Type	Required	Constraints
`query`	string	Yes	1 to 1,000 characters.
`limit`	integer	No	1 to 50. Default: 5.
`filters`	object	No	Optional metadata filters (below).

`filters` **object.** The source lists these fields without stating which are individually required.

Filter field	Type	Constraints
`destination`	string	None stated beyond type.
`travel_type`	string	None stated beyond type.
`min_rating`	number	From 0 to 5. Cannot exceed `max_rating`.
`max_rating`	number	From 0 to 5. Cannot be less than `min_rating`.

RESPONSE FIELDS

JSON object with `query`, `preferences`, `destinations`, `activities`, `evidence`, and `limitations`.

Field	Type	Description
`query`	string	The submitted query.
`preferences`	object	Classified query preferences (below).
`destinations`	array of objects	Ranked destination summaries (below).
`activities`	array of objects	Ranked activity summaries (below).
`evidence`	array of objects	Same result schema as `/api/v1/search` (see Section 3.2).
`limitations`	array of strings	Limitations that apply to the response.

`preferences` object.

Field	Type (from samples)	Description
`interests`	array of strings	Interests identified in the query.
`destination`	string	Destination identified in the query.
`travel_type`	string	Travel type identified in the query.
`duration_days`	integer or null	Sample value: `null` (recommendations), `5` (feasibility).
`budget_amount`	null in samples	Type not shown in the samples.
`confidence`	number	**Illustrative.** May vary with the configured dataset and embedding backend.

Destination object (`destinations[]`).

Field	Type (from sample)	Description
`destination`	string	Destination name.
`average_rating`	number	Average rating.
`evidence_count`	integer	Count of evidence records.
`evidence_ids`	array of strings	`experience_id` values of the evidence.

Activity object (`activities[]`).

Field	Type (from sample)	Description
`category`	string	Activity category.
`average_rating`	number	Average rating.
`evidence_count`	integer	Count of evidence records.

> **Illustrative:** The record fields in the sample reflect the supplied sample row. Scores, confidence, and ranks are illustrative and may vary with the configured dataset and embedding backend.

SAMPLE REQUEST

```
{
  "query": "beaches and restaurants in Lisbon, Portugal for a couple",
  "limit": 1,
  "filters": {
    "destination": "Lisbon, Portugal",
    "travel_type": "Couple",
    "min_rating": 2.24,
    "max_rating": 2.24
  }
}
```

SAMPLE RESPONSE (HTTP 200)


```
{
  "query": "beaches and restaurants in Lisbon, Portugal for a couple",
  "preferences": {
    "interests": ["beaches", "restaurants"],
    "destination": "Lisbon, Portugal",
    "travel_type": "Couple",
    "duration_days": null,
    "budget_amount": null,
    "confidence": 0.74
  },
  "destinations": [
    {
      "destination": "Lisbon, Portugal",
      "average_rating": 2.24,
      "evidence_count": 1,
      "evidence_ids": ["User 24"]
    }
  ],
  "activities": [
    {"category": "beaches", "average_rating": 5.0, "evidence_count": 1},
    {"category": "restaurants", "average_rating": 1.71, "evidence_count": 1}
  ],
  "evidence": [
    {
      "experience_id": "User 24",
      "destination": "Lisbon, Portugal",
      "categories_visited": [
        "resorts", "beaches", "parks", "theatres", "museums", "malls",
        "zoos", "restaurants", "pubs/bars", "local services",
        "burger/pizza shops", "hotels/other lodgings", "juice bars",
        "art galleries", "dance clubs"
      ],
      "overall_rating": 2.24,
      "review_text": "Had a so-so time in Lisbon. Visited as a couple. I was let down by the resorts.
The beaches were the highlight of the trip. Only recommended with low expectations.",
      "travel_type": "Couple",
      "visit_date": "2026-02-24",
      "score": 0.0328,
      "keyword_rank": 1,
      "semantic_rank": 1,
      "synthetic_enrichment": true
    }
  ],
  "limitations": [
    "Enriched review text and trip metadata are synthetic sample data, not verified traveler reviews."
  ]
}
```

3.4 POST /api/v1/itinerary/feasibility

Method	Path	Success status
POST	`/api/v1/itinerary/feasibility`	200

PURPOSE

Return recommendations with a data-coverage assessment for requested itinerary constraints. This endpoint does not calculate actual costs or activity durations because those fields are not in the dataset.

REQUEST

The recommendation request fields plus optional `days` and `budget`.

Field	Type	Required	Constraints
`query`	string	Yes	1 to 1,000 characters.
`limit`	integer	No	1 to 50. Default: 5.
`filters`	object	No	Same filter fields as in Section 3.2.
`days`	integer	No	1 to 30.
`budget`	number	No	Positive number.

RESPONSE FIELDS

JSON object with `status`, `days_requested`, `budget_requested`, `recommendations`, and `limitations`.

Field	Type (from sample)	Description
`status`	string	Data-coverage status (values below).
`days_requested`	integer	Requested number of days (sample: `5`).
`budget_requested`	number	Requested budget (sample: `900.0`).
`recommendations`	object	The recommendation response schema (Section 3.3).
`limitations`	array of strings	Limitations that apply to the response.

`status` values.

Value	When
`insufficient_data`	A budget or duration constraint is provided, because feasibility cannot be established from the available data.
`historical_evidence_only`	No budget or duration constraint is provided.

> **Illustrative:** Confidence, scores, and ranks in the sample are examples and may vary with the configured dataset and embedding backend.

SAMPLE REQUEST


```
{
  "query": "5-day beach trip with local food in Lisbon, Portugal for a couple",
  "limit": 1,
  "days": 5,
  "budget": 900,
  "filters": {
    "destination": "Lisbon, Portugal",
    "travel_type": "Couple",
    "min_rating": 2.24,
    "max_rating": 2.24
  }
}
```

SAMPLE RESPONSE (HTTP 200)


```
{
  "status": "insufficient_data",
  "days_requested": 5,
  "budget_requested": 900.0,
  "recommendations": {
    "query": "5-day beach trip with local food in Lisbon, Portugal for a couple",
    "preferences": {
      "interests": ["beaches", "restaurants"],
      "destination": "Lisbon, Portugal",
      "travel_type": "Couple",
      "duration_days": 5,
      "budget_amount": null,
      "confidence": 0.87
    },
    "destinations": [
      {
        "destination": "Lisbon, Portugal",
        "average_rating": 2.24,
        "evidence_count": 1,
        "evidence_ids": ["User 24"]
      }
    ],
    "activities": [
      {"category": "beaches", "average_rating": 5.0, "evidence_count": 1},
      {"category": "restaurants", "average_rating": 1.71, "evidence_count": 1}
    ],
    "evidence": [
      {
        "experience_id": "User 24",
        "destination": "Lisbon, Portugal",
        "categories_visited": ["resorts", "beaches", "parks", "theatres", "museums", "malls", "zoos",
"restaurants", "pubs/bars", "local services", "burger/pizza shops", "hotels/other lodgings", "juice bars", "art
galleries", "dance clubs"],
        "overall_rating": 2.24,
        "review_text": "Had a so-so time in Lisbon. Visited as a couple. I was let down by the resorts. The
beaches were the highlight of the trip. Only recommended with low expectations.",
        "travel_type": "Couple",
        "visit_date": "2026-02-24",
        "score": 0.0328,
        "keyword_rank": 1,
        "semantic_rank": 1,
        "synthetic_enrichment": true
      }
    ],
    "limitations": [
      "Enriched review text and trip metadata are synthetic sample data, not verified traveler reviews.",
      "Duration feasibility cannot be verified because the dataset contains no activity durations."
    ]
  },
  "limitations": [
    "Enriched review text and trip metadata are synthetic sample data, not verified traveler reviews.",
    "Duration feasibility cannot be verified because the dataset contains no activity durations.",
    "Budget feasibility cannot be verified because the dataset contains no prices."
  ]
}
```

3.5 POST /api/v1/feedback

Method	Path	Success status
POST	`/api/v1/feedback`	201

PURPOSE

Save feedback in the local SQLite store. Feedback is captured but does not currently change retrieval ranking.

REQUEST

JSON object.

Field	Type	Required	Constraints
`experience_id`	string	Yes	1 to 100 characters.
`rating`	integer	Yes	1 to 5.
`comment`	string	No	Up to 1,000 characters. Defaults to an empty string.

RESPONSE FIELDS

JSON object.

Field	Type	Description
`feedback_id`	integer	Identifier of the saved feedback (sample: `1`).
`status`	string	Result status (sample: `"recorded"`).

SAMPLE REQUEST

```
{
  "experience_id": "User 24",
  "rating": 5,
  "comment": "Useful match"
}
```

SAMPLE RESPONSE (HTTP 201)

```
{
  "feedback_id": 1,
  "status": "recorded"
}
```

4. Errors and Status Codes

4.1 Error responses

HTTP status	Meaning	Response shape
422	Invalid JSON fields, blank/oversized query, invalid rating bounds, or invalid feedback.	`{"detail":[...]}`
503	Recommendation service or feedback store is not ready.	`{"detail":"Travel recommendation service is not ready"}` or feedback-store equivalent.

4.2 Success statuses documented per endpoint

Endpoint	Status
`GET /health`	200
`POST /api/v1/search`	200 (including a valid search with no matches: `count: 0` and an empty `results` array)
`POST /api/v1/recommendations`	200
`POST /api/v1/itinerary/feasibility`	200
`POST /api/v1/feedback`	201

All successful endpoint responses use JSON. See `/docs` for the generated OpenAPI schema and interactive request runner.